Yu-Chen Hsieh

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Education

National Yang Ming Chiao Tung University (NYCU), Master of Neuroscience — Sep 2018 – Jun 2020

- GPA: 3.96 / 4.3 (Transcript of record)
- **Coursework:** Cognitive Neuroscience, Fundamentals of neural signal analysis, Advanced Statistics in Psychology

National Tsing Hua University (NTHU), Bachelor of Life Science — Sep 2015 – Jun 2018

- GPA: 3.85 / 4.3 (Transcript of record)
- **Coursework:** Neuroscience, Neuropsychology, Neurobiology

Current Affiliation

Research Assistant, Academia Sinica, Research Center for information technology innovation — May 2026 - Now

- Brain-computer interface, AR/VR
- Human EEG experimental Procedures
- Electroencephalography (EEG) signal analysis

Medical Device Associate Reviewer, Taiwan Food and Drug Administration (TFDA) — Nov 2022 - Feb 2025

- Registration of Electrical Medical Devices
- Amendment Registration of Electrical Medical Devices
- Contract Manufacturing Change Registration

Research Assistant, Academia Sinica, Taiwan Mouse Clinical Pathology Core Lab — Mar 2021 - Nov 2022

- Microscopy Techniques
- Biochemical and Molecular Biological Techniques
- Animal Experimental Procedures

Research Experience

Development and validation of a novel 32-channel rodent EEG system — Jul 2018 - Nov 2020

- Integrated EEG pipeline: Designed custom electrodes, performed impedance correction, recorded ERPs, and analyzed data in rodent models.
- Differentiated neural signatures: Discovered distinct ERP patterns for passive vs. reward-modulated visual / auditory stimuli.
- Captured state dynamics: Tracked arousal-dependent oscillatory changes continuously throughout recordings.
- Validated implant stability: Maintained a low, stable impedance for $>$ 6 months of behavioral record.

The effect of oscillatory activity on axon regeneration of retinal ganglion cells in *rd1* mice — Feb 2016 – Jul 2018

- Demonstrated that electrical oscillations enhance axonal regeneration in cultured retinal tissue.
- Validated ganglion cell electrical activity using multielectrode array (MEA), modulated by pharmacological agents.

Honors and Awards

Fellow, Taiwan Global Pathfinders Initiative, Project No.I-8-7 (Certification) — 2026

Outstanding Award, Student Presentation Competition, Taiwan Society of Cognitive Neuroscience — 2020

Best Thesis Award (1st place), Institute of Neuroscience, NYCU 2020

Outstanding Thesis Scholarship, Master's Program, NYCU 2020

Best Popularity Award, Poster Competition, Institute of Neuroscience, NYCU 2020

Best Popularity Award, Poster Competition, College of Life Science and Medicine, NTHU 2018

Academic Excellence Award, Junior Fall and Junior Spring, NTHU 2017

Key Techniques

Languages: Mandarin, English

Data Analysis and Programming:

- MATLAB (ERP data processing, EEG signal analysis)
- EEGlab toolbox
- Python (Data visualization, Basic scripting)

Laboratory Techniques:

- Rodent Microsurgery (stereotaxic implantation)
- Microscopy Techniques
- Biochemical and Molecular Biological Techniques
- Animal Experimental Procedures

Hardware:

- Basic electronics and Soldering (PCB assembly and rework)

Reference

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